Name of Student:

Sch. No.:

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

DEPARTMENT OF ELECTRICAL ENGINEERING

END TERM EXAMINATION May.-2025, B. Tech. IV Semester Electrical

Sub: Generation of Electrical Power (E-224)

Time: 3.00 Hours **Max. Marks: 50**

Note: Assume missing data if necessary. Both parts of a question should be solved at ONE place only.

I.	a)	With schematic diagram explain working principle of power generation from (i) Solar Thermal (ii) Wind. Also discuss their merits and demerits.	05	CO2
	b)	The yearly load duration curve of a certain power station can be considered as a straight line from 300 MW to 80 MW. Power is supplied with one generator unit of 200 MW capacity and two units of 100 MW capacities each. Determine (i) Installed capacity (ii) Load factor (iii) Plant factor (iv) Max. Demand (v) utilization factor	05	CO3
II	a)	With neat diagram explain the Electricity market models in deregulated power systems.	05	CO1
	b)	A factory has an induction motor having a maximum demand of 800 kW at 0.707 pf lagging. The consumer is charged at Rs 80 per kVA of the maximum demand per year. The increase in load is met by installing a synchronous motor of 200 hp and efficiency of 0.99. If the synchronous motor works on full load and at 0.8 pf leading find the difference in annual fixed charges of consumer.	05	CO3
III.	a)	Draw the water-steam flow diagram for a boiler turbine unit and explain function of each component in brief.	05	CO2
	b)	With neat sketch, discuss the functions of following auxiliaries in thermal power stations (i) Economizer (ii) Air pre-heater (iii) Super heater (iv) Electro Static Precipitator (v) Station transformer.	05	CO2
IV	a)	With the help of a neat sketch explain the working principle of advanced gas cooled reactor used in a nuclear power plant. How MAGNOX reactor differs from advanced gas cooled reactor.	05	CO2

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	b)	How can hydro plants are classified according to (i) water flow regulation (ii) head (iii) load	05	CO1
V	a)	Analyze the solution of Coordination equation using (i) Analytical Method (ii) Graphical Method	05	CO3
	b)	Two units of a thermal station have each the following cost characteristics $C = 5000 + 450P + 0.5P^2 \ Rs./hr$ Due to an instrumental error the cost characteristics of first unit is in error by +2% and that of the second unit by -2% at the time of scheduling. Find the extra operating cost due to erroneous scheduling. Total load is 200MW	05	CO3

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